

Gravitational lensing is not independent evidence for dark matter: the no-slip identity in a cosmological-constant acceleration-scale framework

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Gravitational lensing is routinely presented as direct, model-independent evidence for particle dark matter—the Bullet cluster’s mass–gas offset, cluster strong+weak-lensing masses, the galaxy–galaxy-lensing radial acceleration relation, and strong-lens Einstein radii. We show that within a framework in which the modified-dynamics acceleration scale is set by the cosmological constant, $a_0 = c^2\sqrt{\Lambda/32\pi} = 9.36 \times 10^{-11} \text{ m s}^{-2}$, realised covariantly by the Aether-Scalar-Tensor (AeST) theory, this evidentiary status collapses. AeST has no gravitational slip: the two weak-field potentials are equal, $\Phi = \Psi$, so the lensing mass equals the dynamical mass for any source. Four of the standard lensing-for-dark-matter arguments therefore reduce to a single statement—that clusters carry a residual between dynamical and baryonic mass—rather than constituting four independent proofs. We verify the identity from the AeST field equations and the primary literature, audit the acceleration-scale footing of the measurements (cluster lensing masses are pure general-relativity+dark-matter deductions with no a_0 ; the framework uses its own least-favourable a_0), and report the standing both ways: the galaxy–galaxy-lensing relation passes; the cluster residual ($\eta \simeq 2.3$) and the cosmic-microwave-background third peak remain genuinely independent, conceded losses; the once-headline $8.6\text{--}9.2\sigma$ weak-lensing morphology split is now contested by a re-analysis of the same data. Lensing, in this framework, is not a dark-matter detector but a second view of the one shared cluster residual.

I. THE CLAIM UNDER EXAMINATION

Among the empirical pillars of cold dark matter, gravitational lensing occupies a special rhetorical place: it is said to weigh mass directly, through the deflection of light, independently of any assumption about dynamical equilibrium. The canonical statements are (a) the Bullet cluster 1E 0657–558, whose weak- and strong-lensing convergence peaks are spatially offset from the dominant baryonic component (the X-ray gas) and track the collisionless galaxies [1]; (b) cluster strong+weak-lensing masses that exceed the baryonic mass by a factor ~ 2 ; (c) the galaxy–galaxy-lensing radial acceleration relation (RAR), which extends the rotation-curve RAR to accelerations below those reached kinematically [3]; and (d) strong-lens Einstein radii and time delays. Each is presented as an *independent* confirmation that gravitating mass exceeds visible mass.

The premise behind “independent” is that lensing and dynamics probe different things. In general relativity with a dark halo they do not: both follow the same total mass. But for *modified gravity* the premise is non-trivial, because the original non-relativistic Milgromian dynamics (MOND [2]) famously *under*-predicted lensing—a real historical failure. The question this paper settles, for the present framework, is whether its *relativistic* completion repairs that failure in a way that also dissolves the “independent evidence” claim.

II. THE FRAMEWORK AND ITS RELATIVISTIC LENSING

The framework posits that the MOND acceleration scale is not a free constant but is fixed by the cosmological constant,

$$a_0 = c^2\sqrt{\Lambda/32\pi} = \frac{c}{2}\sqrt{G\rho_{\text{DE}}} = cH_\Lambda/Z, \quad Z = \sqrt{\frac{32\pi}{3}}, \quad (1)$$

giving $a_0 = 9.36 \times 10^{-11} \text{ m s}^{-2}$ for the measured Λ ; the coefficient $Z = 5.79$ is an unforced posit, flagged here and not asserted as derived. The non-relativistic dynamics follow the de Sitter–Unruh interpolation $g_{\text{obs}} = \sqrt{g_{\text{bar}}^2 + g_{\text{bar}}a_0}$. The covariant realisation is the Aether-Scalar-Tensor theory (AeST) [4], a Lorentz-violating scalar–vector–tensor theory whose unit-timelike vector field is the device that fixes the lensing.

a. No gravitational slip. Write the perturbed Friedmann–Robertson–Walker metric in conformal Newtonian gauge as $ds^2 = -(1 + 2\Phi)dt^2 + (1 - 2\Psi)d\mathbf{x}^2$. Dynamics (non-relativistic test bodies) respond to Φ ; light deflection responds to $\Phi + \Psi$. In general MOND-like theories these can differ (a “slip” $\Phi \neq \Psi$), which is exactly how the old non-relativistic MOND under-lensed. AeST is constructed so that they do not. The quasistatic solutions of AeST [5] state it directly: “the two standard weak-field metric potentials Φ and Ψ are equal ... this ensures that the lensing mass is equal to the dynamical mass ... for any source configuration where the MOND limit of AeST successfully replaces dark matter, the gravitational lensing signal will also be as if dark matter is present.” Equivalently, a single effective potential enters both metric components [6]; the unit-timelike vector

TABLE I. The lensing-for-dark-matter arguments under the no-slip identity. “Reduces” = not independent of the cluster dynamical residual; “independent” = untouched by Eq. (2).

Argument	Status	Note
Bullet mass–gas offset	reduces	collisionless residual
Cluster SL+WL $\sim 2\times$	reduces	= dynamical η
Galaxy–galaxy lensing RAR	reduces	passes (below)
Strong-lens Einstein radii	reduces	same potential
CMB third peak / $P(k)$	independent	loss, a_0 -free
Lensed-quasar substructure	independent	open

sources Ψ alongside Φ , the Sanders mechanism. Thus

$$\Phi = \Psi \implies M_{\text{lens}} = M_{\text{dyn}} \quad (\text{for any source}). \quad (2)$$

This is a field-equation identity, not a fit. Its only caveat is in the far field: beyond a transition radius the AeST potentials enter an oscillatory regime [5], and weak-lensing tests at the largest radii constrain the amplitude [6]; neither introduces a slip at the radii relevant below.

III. THE REFRAME

Equation (2) removes the logical independence of the lensing arguments. If lensing mass equals dynamical mass identically, then a lensing measurement that finds “more mass than baryons” finds *the same* excess the dynamics already show—not a second, corroborating excess. Table I sorts the standard arguments accordingly.

Four arguments collapse into one: *do clusters carry a dynamical residual?* They do— $\eta(R_{500}) \simeq 2.3$ on the eROSITA eRASS1 sample of 9830 clusters [7]—and that residual is the well-known, shared liability of all modified-dynamics theories, not a new lensing discovery. Two arguments survive as genuinely independent: the matter power spectrum encoded in the cosmic-microwave-background acoustic peaks (where a_0 is absent from the linear physics, so the third-peak height is a real, conceded loss); and the abundance of 10^7 – $10^9 M_\odot$ substructure inferred from lensed-quasar flux-ratio anomalies, which probes clumpiness rather than the smooth potential and is open in both directions.

IV. THE ACCELERATION-SCALE FOOTING

A recurring concern is whether the comparison silently uses a non-framework a_0 . We audited it. Cluster weak-lensing masses (eRASS1, DES-Y3, HSC) are computed in pure general relativity with a dark halo and contain *no* a_0 : the inferred “mass needed” is simply the lensing-deduced enclosed mass exceeding the baryons. The framework comparison then uses Eq. (1), $a_0 =$

9.36×10^{-11} , which is the *least*-favourable choice, because in the deep regime $\eta \propto a_0^{-1/2}$: the canonical MOND value 1.2×10^{-10} would make the residual appear $\sim 13\%$ *smaller*. On the eRASS1 data we find $\eta(R_{500}) = 2.33$ on the framework’s own interpolation, versus 1.92 at the canonical value—so the literature, which uses the canonical value, if anything *understates* the framework’s residual. The galaxy–galaxy-lensing estimator is itself a_0 -free ($g_{\text{obs}} \propto \Delta\Sigma$), and the morphology split below is a_0 - and interpolation-independent by construction. There is no footing error inflating any lensing loss; the only mis-footings we found ran the other way (a galaxy-lensing amplitude and a cluster residual once displayed at the canonical a_0 , flattering the framework), and correcting them makes its numbers slightly worse—which we adopt.

V. THE BULLET CLUSTER

The Bullet is the iconic case. Under Eq. (2) its lensing peaks lie where the total gravitating mass is, which in modified gravity is sourced by all matter; the peaks sitting on the collisionless galaxies rather than the shocked gas requires that the cluster residual be *collisionless* and galaxy-tracking—cold gas, stellar remnants, or relic neutrinos—which is the standard residual, offset from the gas because the gas alone is collisional. Recent consistent-QUMOND lens modelling reproduces the convergence-on-galaxies morphology with a residual of order the cluster mass [8]; we treat this as imported rather than derived. The Bullet is thus a mild, shared liability, not a clean falsification of modified gravity—and its $\sim 4000 \text{ km s}^{-1}$ collision velocity remains a long-standing tension for ΛCDM pairwise velocities that survives the 2025 reclassification of the system as a minor merger [9]. The complementary external-field prediction for the dark-matter-deficient galaxy NGC 1052-DF2 ($\sigma \sim 14 \text{ km s}^{-1}$) is a clean modified-dynamics success.

VI. THE MORPHOLOGY SPLIT

One lensing tension does *not* reduce: the galaxy–galaxy-lensing RAR split by morphology. In the released KiDS catalogue, early-type galaxies lie $+0.26$ dex above late types at fixed baryonic acceleration—an 8.6 – 9.2σ contrast [3]. It is a_0 - and interpolation-independent: a property-blind force law predicts a single relation, and we confirm the early-minus-late model difference cancels identically for every a_0 and interpolation. An intrinsic split of this size would falsify the law. It is, however, now *contested*: a re-analysis of the same data by the RAR’s own authors, applying a stricter isolation cut, finds the early/late deviation falling from $\sim 7.5\sigma$ to ~ 0 between isolation radii of 2.7 and 4.0 Mpc (the original used 3 Mpc), attributing the residual to early-type two-halo and satellite contamination from morphology–density clustering plus a mild stellar-initial-mass-function offset [10].

This relocates the split to a selection systematic rather than the intrinsic type-dependence the law forbids; but it rests on a smaller, noisier sample and does not erase the released-catalogue contrast. We present it as an open selection-versus-intrinsic question, decidable by the deeper, better-isolated samples of Euclid and Rubin this decade.

VII. WHAT LENSING DOES AND DOES NOT ESTABLISH

The honest ledger, both ways. *Resolved*: the no-slip identity is real and removes the logical independence of the Bullet, cluster strong+weak lensing, the bulk galaxy-lensing RAR, and Einstein radii from the dark-matter case; the galaxy–galaxy-lensing RAR passes at the framework’s own a_0 (a soft pass, inside the data scatter [3]), curing the historical under-lensing of non-relativistic MOND. *Conceded, genuinely independent*: the cluster residual $\eta \simeq 2.3$ (real, confirmed by lensing independently of hydrostatic bias, shared with all MOND, and uncured here—the AeST scalar mass term is a candidate but is wrong-signed at clusters and squeezed by galaxy lensing [6, 11]); and the CMB third peak. *Contested*: the morphology split. *Open*: lensed-quasar substructure. Lensing is therefore not, in this framework, an independent detector of a dark particle; it is a second window on the one shared cluster residual, with the framework’s largest specifically lensing tension—the morphology split—now in dispute.

VIII. CONCLUSION

“Gravitational lensing proves dark matter” is, for a relativistic modified-dynamics theory with no gravitational slip, a statement that double-counts. The lensing mass equals the dynamical mass by construction, so the deflection-of-light arguments restate the cluster dynamical residual rather than corroborate it independently. This does not make the cluster residual disappear—it is real, and conceded—nor does it touch the genuinely independent constraints from the cosmic-microwave-background power spectrum. But it removes lensing from the list of independent proofs of a dark particle and returns the dark-sector question to where the framework already locates it: a single, soft, shared residual at the centres of clusters, and the one falsifiable thing the framework alone predicts, a redshift-declining $a_0(z) \propto \sqrt{\rho_{\text{DE}}(z)}$.

ACKNOWLEDGMENTS

The acceleration-scale footing audit, the eRASS1 cluster residual at the framework’s own a_0 , and the morphology-split cancellation were performed and cross-checked numerically; the no-slip identity is quoted from the primary AeST literature. Both-ways audit ledgers are archived with this work.

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